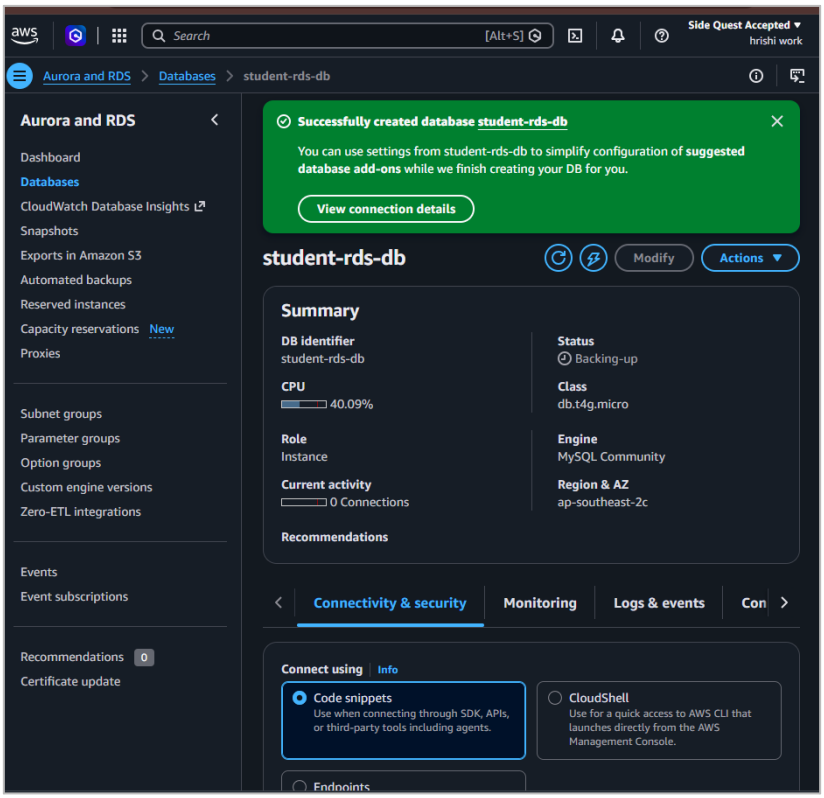
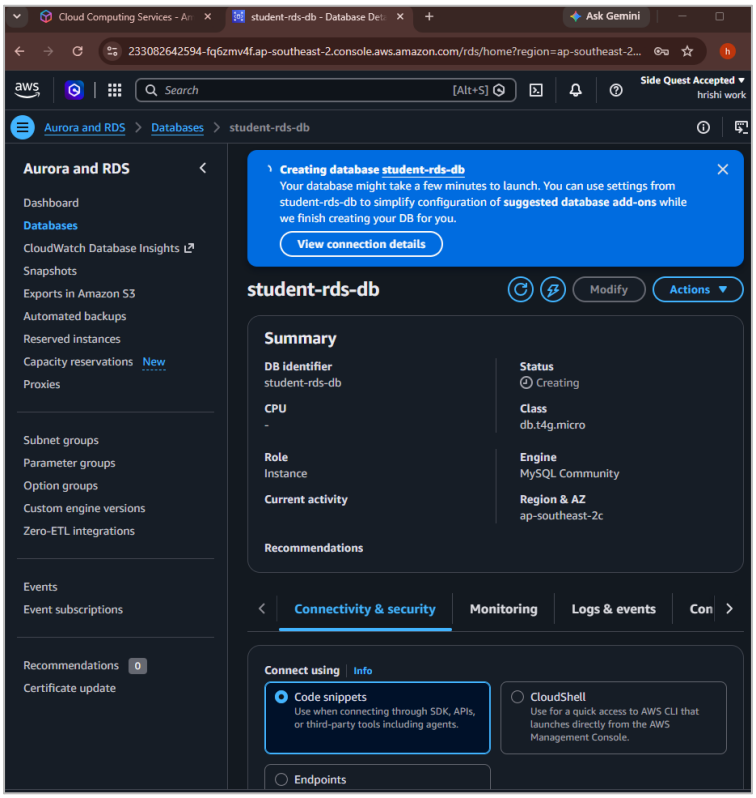
AWS RDS TASKS

Submitted by Hrishinandan P

Task 1: Create an Amazon RDS Database

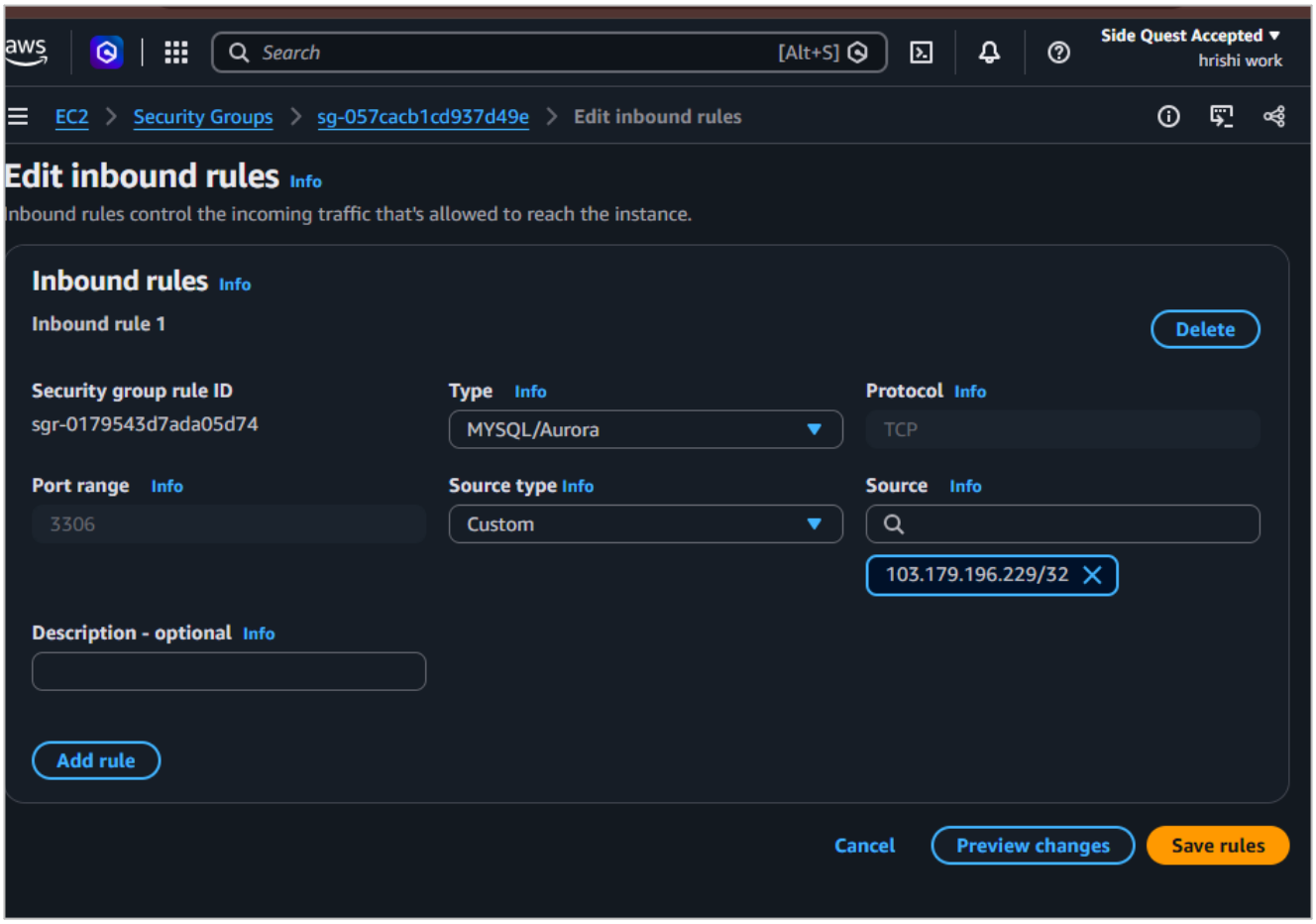
- Launched a MySQL Community RDS instance (**student-rds-db**, class db.t4g.micro) in ap-southeast-2.
- The database was created successfully; the endpoint is available under Connectivity & security.





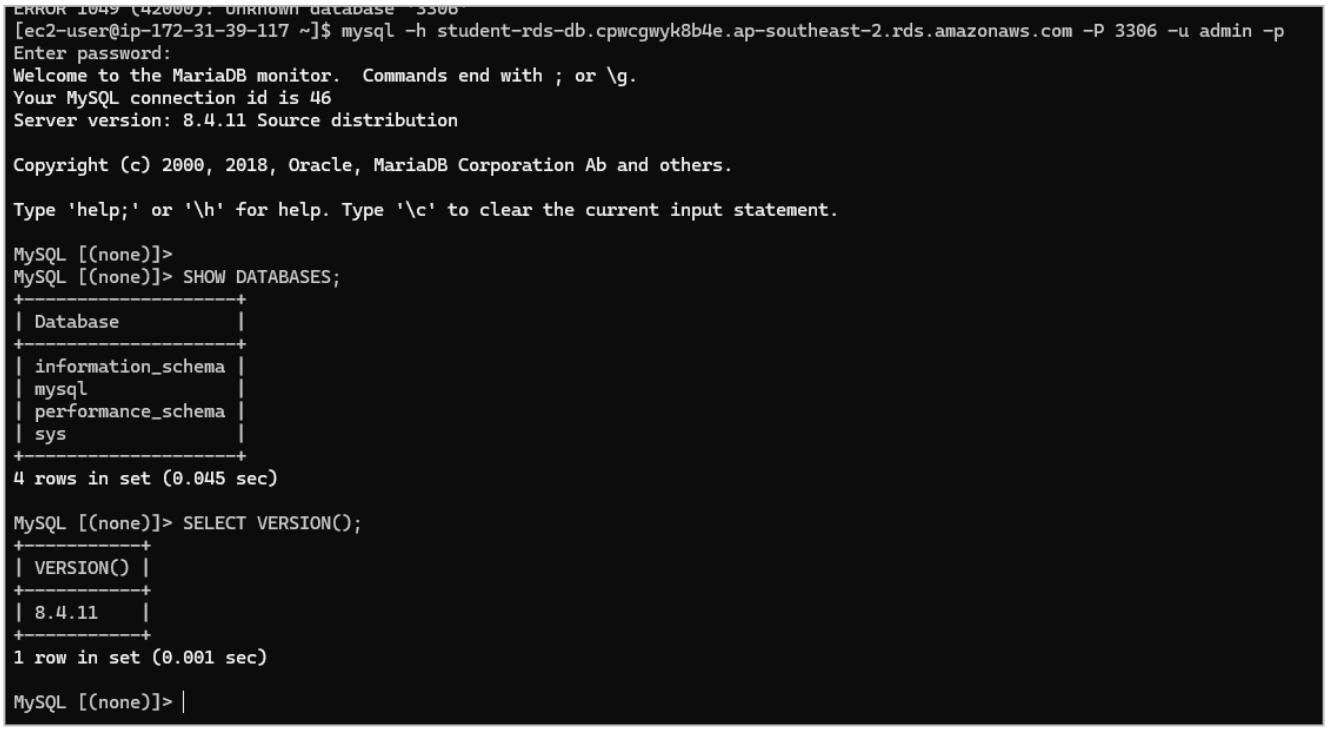
Task 2: Configure Security Group

- Inbound rule allows MySQL/Aurora (TCP 3306) only from my public IP (**103.179.196.229/32**) on security group sg-057cacb1cd937d49e.
- 0.0.0.0/0 is insecure because it lets any IPv4 address on the internet reach the database, exposing it to brute-force logins, unauthorized access and vulnerability exploitation.



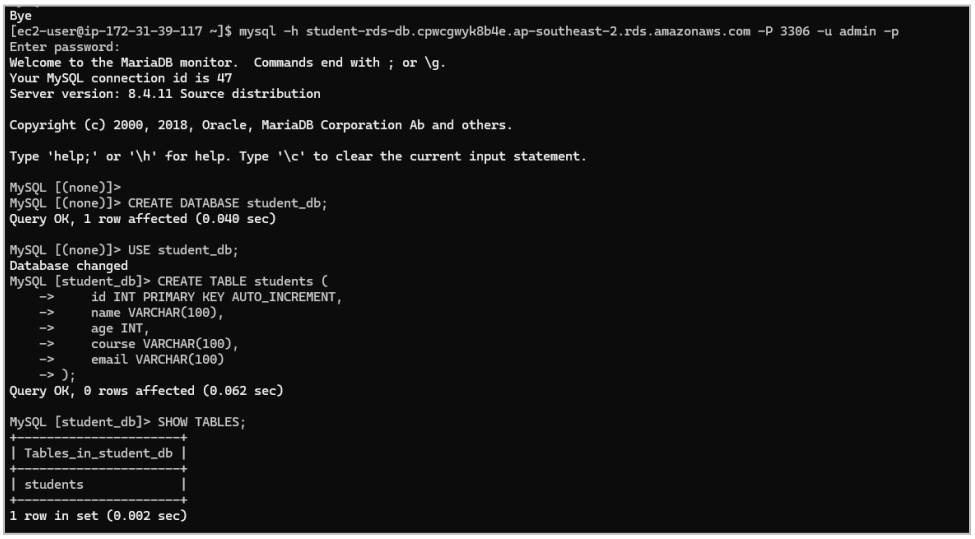
Task 3: Connect to RDS

- Connected to the RDS endpoint from an EC2 instance using the MySQL CLI (mysql -h <endpoint> -P 3306 -u admin -p).
- **SHOW DATABASES** listed the default databases and **SELECT VERSION()** returned 8.4.11.



Task 4: Create Database Objects

- Created database **student_db** and a **students** table (id, name, age, course, email), then inserted 3 records and verified them with SELECT.
- Updated Rahul's age to 24 and deleted the record with id 1; the final SELECT shows 2 remaining rows.



```
Bye
[ec2-user@ip-172-31-39-117 ~]$ mysql -h student-rds-db.cpwcgwyk8b4e.ap-southeast-2.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 47
Server version: 8.4.11 Source distribution

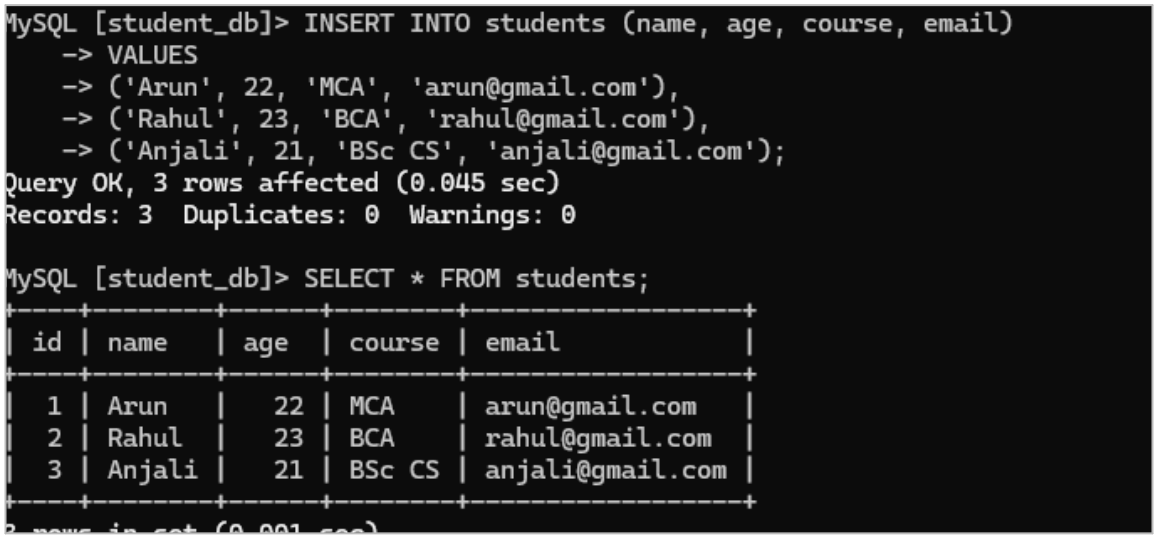
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
MySQL [(none)]> CREATE DATABASE student_db;
Query OK, 1 row affected (0.040 sec)

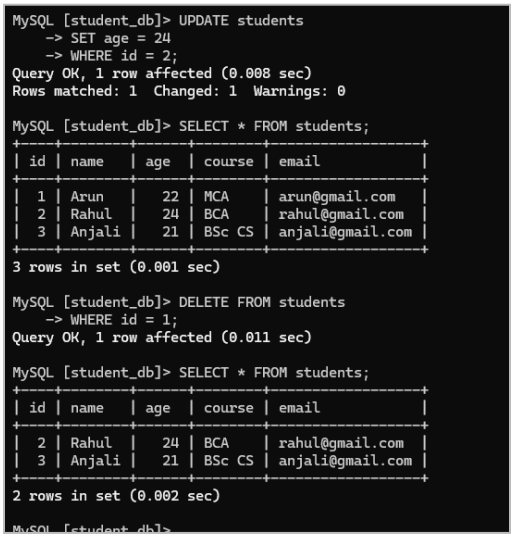
MySQL [(none)]> USE student_db;
Database changed
MySQL [student_db]> CREATE TABLE students (
    ->     id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100),
    ->     age INT,
    ->     course VARCHAR(100),
    ->     email VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.062 sec)

MySQL [student_db]> SHOW TABLES;
+----------------------+
| Tables_in_student_db |
+----------------------+
| students             |
+----------------------+
1 row in set (0.002 sec)
```



```
MySQL [student_db]> INSERT INTO students (name, age, course, email)
    -> VALUES
    -> ('Arun', 22, 'MCA', 'arun@gmail.com'),
    -> ('Rahul', 23, 'BCA', 'rahul@gmail.com'),
    -> ('Anjali', 21, 'BSc CS', 'anjali@gmail.com');
Query OK, 3 rows affected (0.045 sec)
Records: 3  Duplicates: 0  Warnings: 0

MySQL [student_db]> SELECT * FROM students;
+----+--------+------+---------+-------------------+
| id | name   | age  | course  | email             |
+----+--------+------+---------+-------------------+
|  1 | Arun   |   22 | MCA     | arun@gmail.com    |
|  2 | Rahul  |   23 | BCA     | rahul@gmail.com   |
|  3 | Anjali |   21 | BSc CS  | anjali@gmail.com  |
+----+--------+------+---------+-------------------+
```



```
MySQL [student_db]> UPDATE students
    -> SET age = 24
    -> WHERE id = 2;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

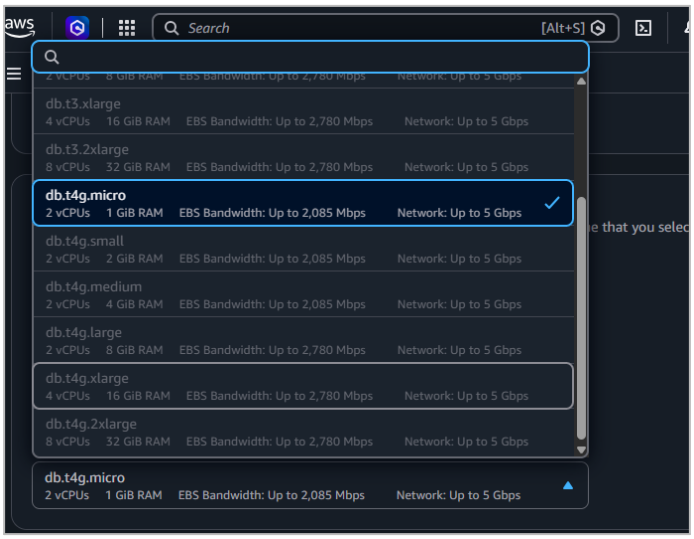
MySQL [student_db]> SELECT * FROM students;
+----+--------+------+---------+-------------------+
| id | name   | age  | course  | email             |
+----+--------+------+---------+-------------------+
|  1 | Arun   |   22 | MCA     | arun@gmail.com    |
|  2 | Rahul  |   24 | BCA     | rahul@gmail.com   |
|  3 | Anjali |   21 | BSc CS  | anjali@gmail.com  |
+----+--------+------+---------+-------------------+
3 rows in set (0.001 sec)

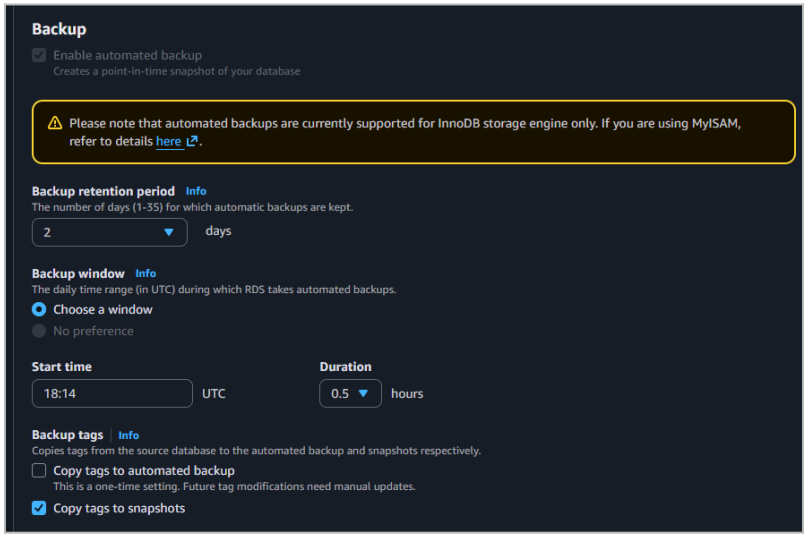
MySQL [student_db]> DELETE FROM students
    -> WHERE id = 1;
Query OK, 1 row affected (0.011 sec)

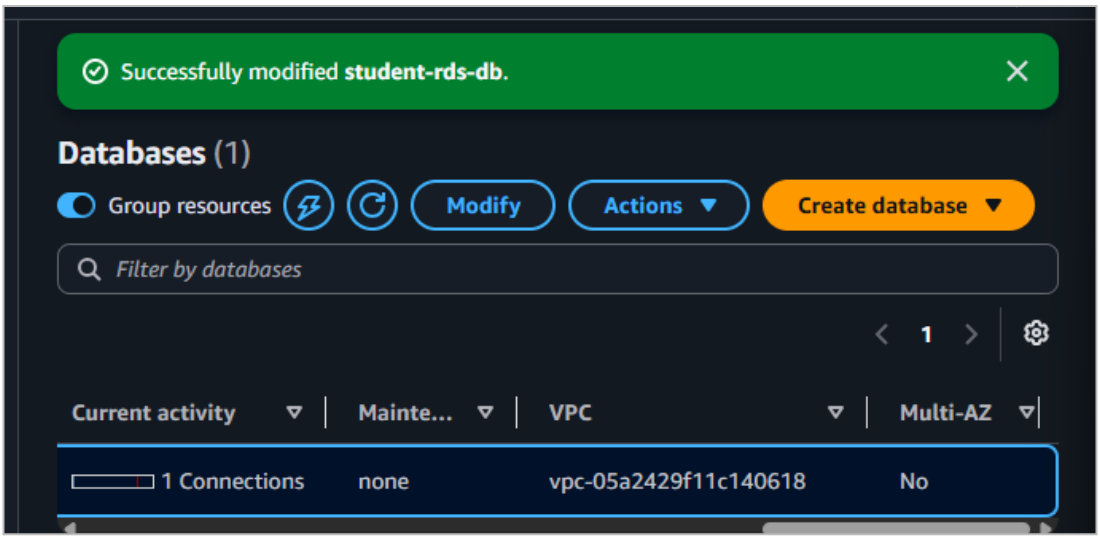
MySQL [student_db]> SELECT * FROM students;
+----+--------+------+---------+-------------------+
| id | name   | age  | course  | email             |
+----+--------+------+---------+-------------------+
|  2 | Rahul  |   24 | BCA     | rahul@gmail.com   |
|  3 | Anjali |   21 | BSc CS  | anjali@gmail.com  |
+----+--------+------+---------+-------------------+
2 rows in set (0.002 sec)
```

Task 5: Modify the RDS Instance

- Opened Modify on **student-rds-db** to review instance class, storage and backup settings, and applied the changes.
- AWS confirmed the modification with a "Successfully modified student-rds-db" message.

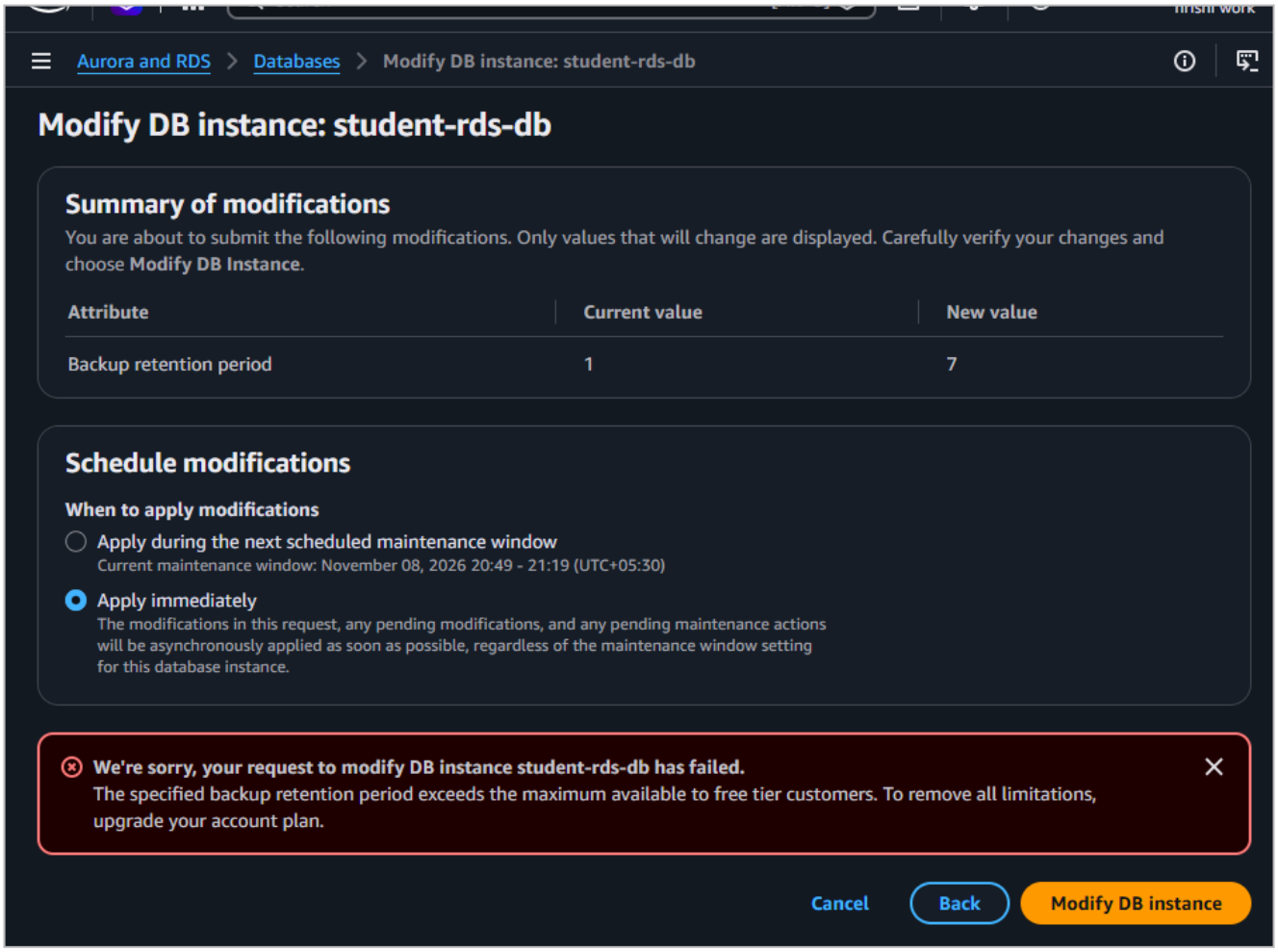






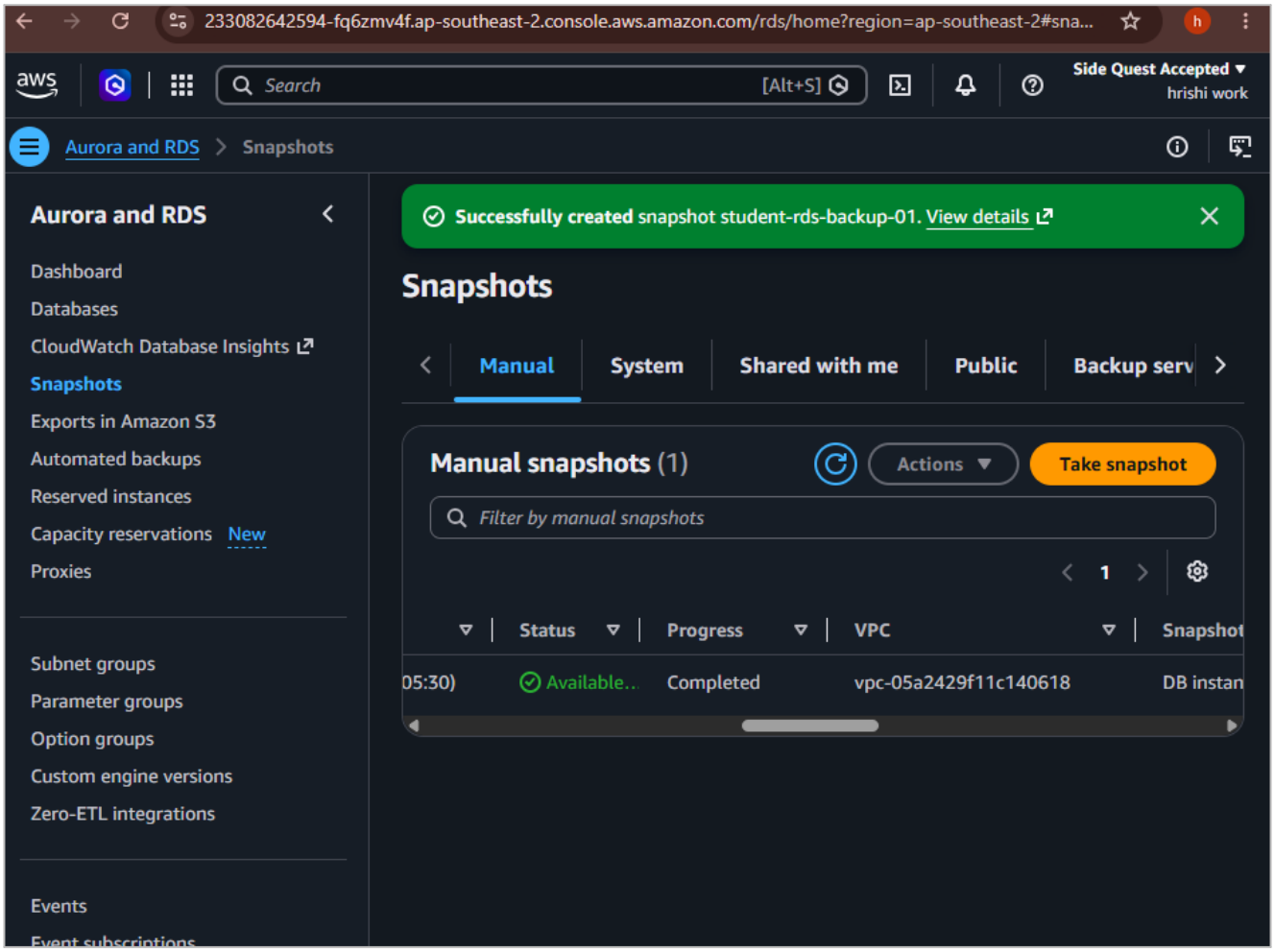
Task 6: Enable Automated Backups

- Automated backups are enabled: RDS takes a daily backup in the backup window and keeps it for the retention period, allowing point-in-time restore.
- A 7-day retention period was requested through Modify DB instance; the free plan limits the maximum retention allowed.



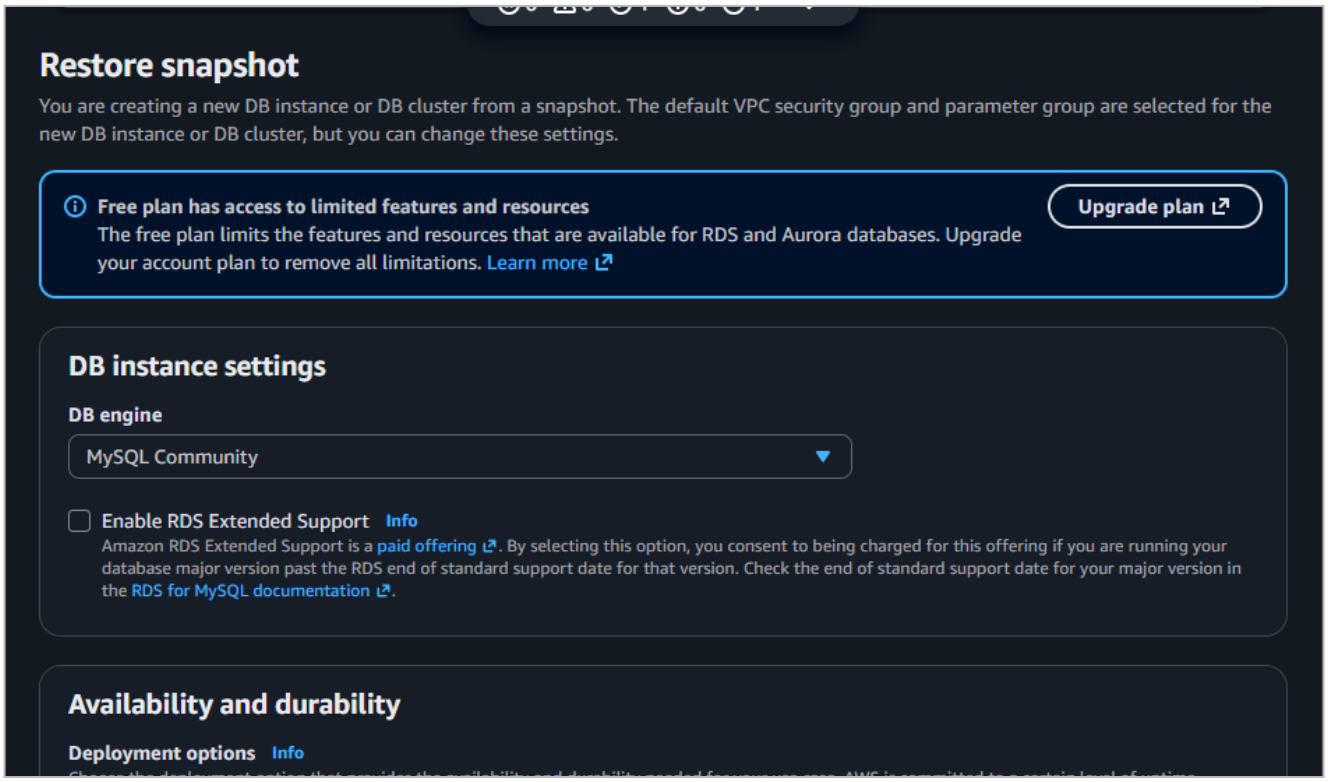
Task 7: Create a Manual Snapshot

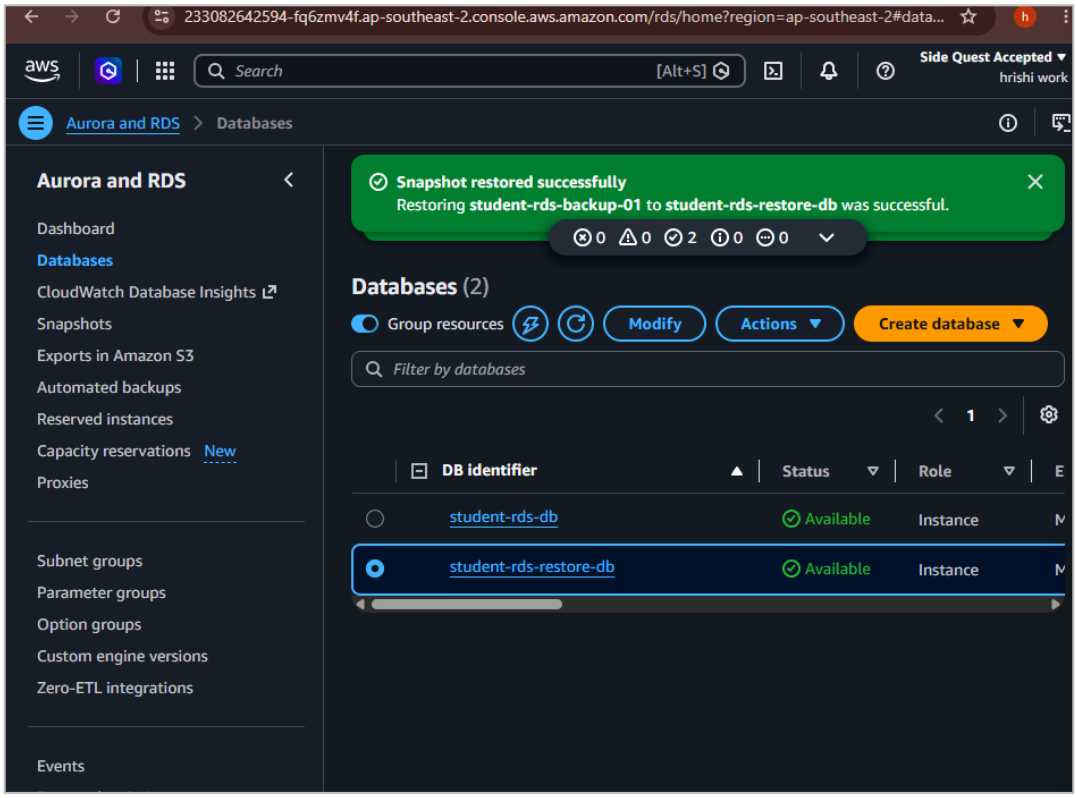
- Created a manual snapshot, **student-rds-backup-01**, of student-rds-db.
- Verified in the Manual snapshots list that its status is Available and progress is Completed.



Task 8: Restore from Snapshot

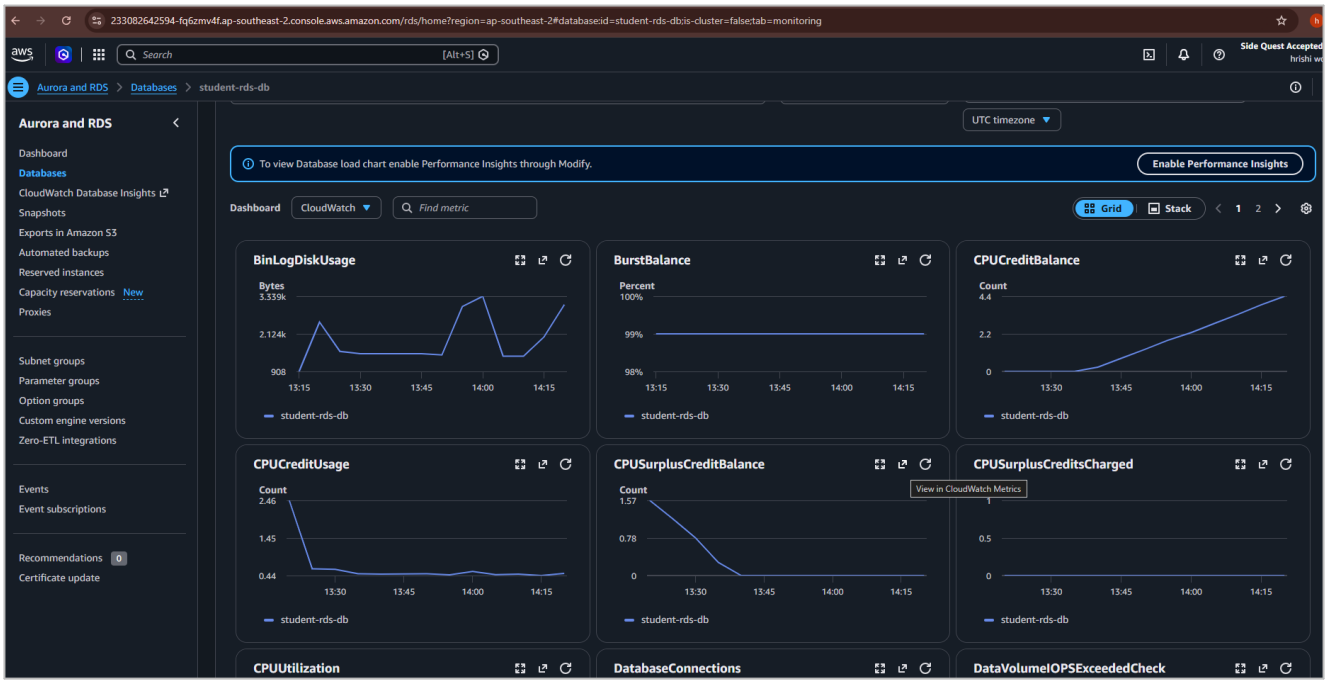
- Restored a new RDS instance, **student-rds-restore-db**, from the manual snapshot student-rds-backup-01.
- The restore completed successfully and both instances show status Available.





Task 9: Monitor Performance

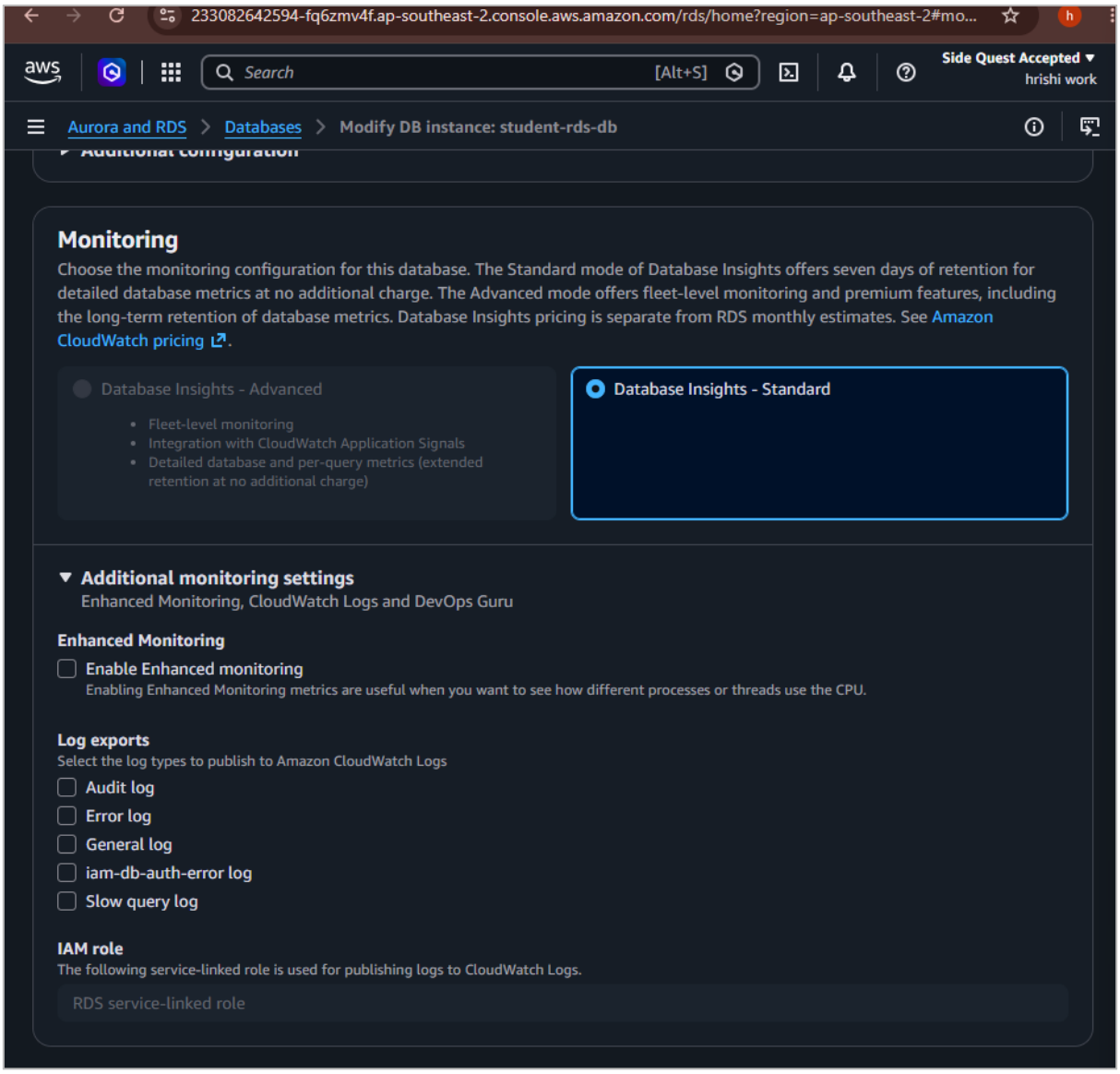
- Opened the Monitoring tab of student-rds-db to view CloudWatch metrics for CPU, memory, storage and IOPS.
- Free storage and memory stayed stable after an initial drop at launch, and IOPS settled to a low steady level.





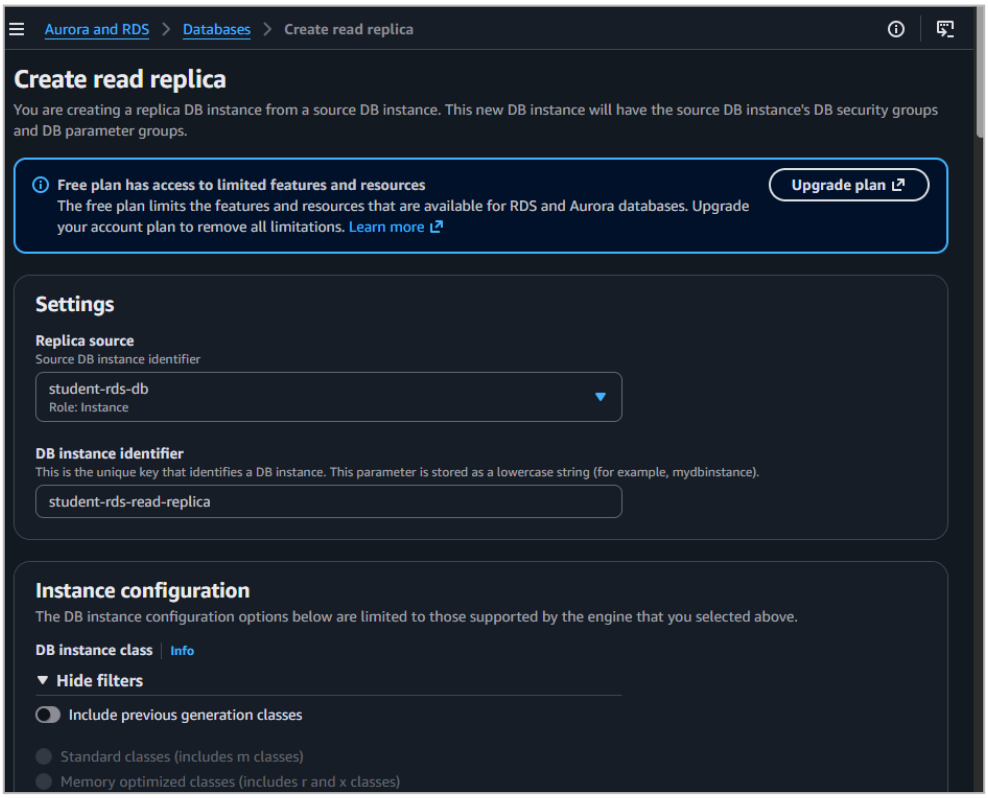
Task 10: Performance Insights

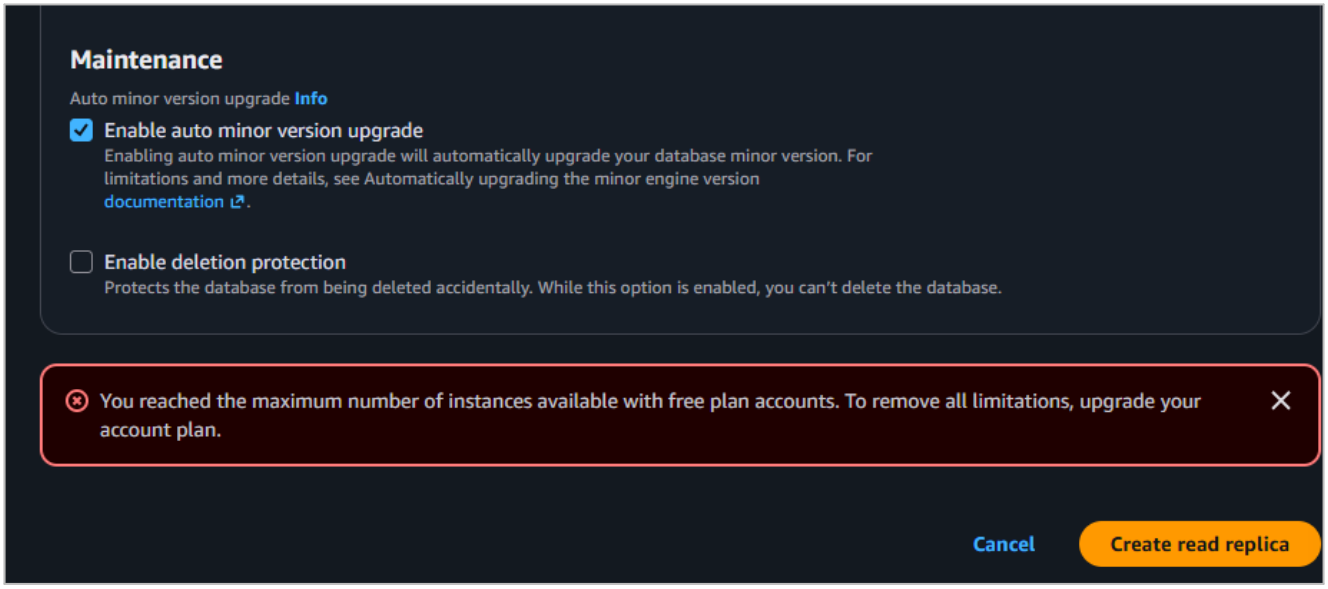
- Configured Database Insights (Standard mode), the current replacement for Performance Insights, on student-rds-db through Modify.
- Standard mode keeps 7 days of detailed database metrics at no additional charge.



Task 11: Read Replica (Optional)

- Opened Create read replica for the source instance **student-rds-db** and set the replica identifier to student-rds-read-replica.
- A read replica is a read-only copy of the source database that uses asynchronous replication to offload read traffic.





Task 12: Cleanup

- Deleted the restored instance **student-rds-restore-db**, creating a final snapshot (student-rds-restore-db-snapshot) before deletion.
- Deleted the manual snapshot student-rds-backup-01 and terminated the client EC2 instance (RDS-Client-EC2).

